1 Finite Dougall tails and a harmonic composition generator

The centered all-resonance continuation gives an exact digamma value for an ordinary subtracted sum. Finite termination is a stronger property. The following classification and generating identity are proved in the accompanying report; compact proofs are included here.

1.1 Rationality and exact termination

Let $N \geq 0$, $\kappa, b, c > 0$, $d_0 = 1 + 2\kappa - b - c > 0$, and $d = d_0 + N$. Put

$$\alpha = (\kappa, b - \kappa, c - \kappa, d - \kappa), \qquad F_N(x) = x \prod_{i=1}^{4} \frac{\Gamma(x + \alpha_i)}{\Gamma(x + 1 - \alpha_i)}.$$

Denominator Gamma factors are entire reciprocal-Gamma factors. The centered expansion has leading power x^{2N-1} and even inverse-power corrections.

Theorem 1.1 (Complete integral-resonance termination locus)**.** *The function F_N is rational if and only if some member of $\{b, c, d\}$ is integral. Its centered expansion terminates if and only if either some member lies in $\{1, \ldots, N\}$, or (b, c, d) is a permutation of $(N + 1, \kappa, \kappa)$. In the second case,*

$$F_N(x) = x^{-1} \prod_{j=1}^{N} (x^2 - (\kappa - j)^2).$$

Proof. A Gamma quotient is rational precisely when its numerator and denominator shift multisets agree modulo $\mathbb{Z}$: otherwise there are infinitely many uncanceled zeros or poles in an arithmetic progression; in the matched case Gamma recurrence gives a rational product. Here this is negation invariance of the four classes α_i. Their total sum is $N + 1$. Negation invariance with integral total forces α_1 to have a distinct partner whose sum with it is integral. For a fixed class 0 or $1/2$, failure of such a partner would leave an odd number of the opposite fixed class and make the total half-integral. The three possible pair sums are b, c, d. Conversely any such integral sum also makes the complementary pair integral. This proves rationality.

Choose an integral member m and rename the others c_*, d_*. If $1 \leq m \leq N$, recurrence gives the polynomial

$$x \prod_{j=1}^{m-1} (x^2 - (\kappa - j)^2) \prod_{r=1}^{N-m} (x^2 - (\kappa - c_* + r)^2).$$

Otherwise choose the smallest integral member and set $L = m - N \geq 1$. The quotient becomes

$$x \frac{\prod_{j=1}^{m-1} (x^2 - (\kappa - j)^2)}{\prod_{r=0}^{L-1} (x^2 - (\kappa - c_* - r)^2)}.$$

Cancellation requires $c_* + r = j$ or $d_* + L - 1 - r = j$ with $1 \leq j < m$. This is impossible: each of c_*, d_* is either nonintegral or at least m. Thus all denominator roots must be zero for a Laurent polynomial. This forces $L = 1$, $c_* = d_* = \kappa$.

Finally, formal finite termination is exact. The quotient satisfies

$$F_N(x + 1)/F_N(x) = \frac{x + 1}{x} \prod_i \frac{x + \alpha_i}{x + 1 - \alpha_i}.$$

If its all-orders expansion is a finite Laurent polynomial A, the rational function $A(x + 1) - R(x)A(x)$ is flat at infinity and hence zero. Thus F_N/A is one-periodic and tends to one on positive real translates. It equals one identically. This excludes an unexamined flat remainder and proves the theorem. □

Zero sum is not termination. For $0 < \kappa < 1$, the analytic implicit function theorem gives an even $s(v)$ with $s(0) = 0$ satisfying

$$\psi(N+1) - \psi(N+1-2s) - \psi(\kappa+s+v) - \psi(\kappa+s-v) + 2\psi(\kappa) = 0.$$

Indeed the s-derivative at zero is $2[\psi'(N+1) - \psi'(\kappa)] \neq 0$ and

$$s''(0) = \frac{\psi''(\kappa)}{\psi'(N+1) - \psi'(\kappa)} > 0.$$

Set $b = \kappa + s(v) + v$, $c = \kappa + s(v) - v$, $d_0 = 1 - 2s(v)$. The established collapsed-remainder formula makes the summed remainder zero, but for small real $v \neq 0$ none of b, c, d is integral. The summand is then not even rational. This is not a converse to the termination theorem.

1.2 An ordinary harmonic resolvent

Define

$$P_n(u) = \frac{(1-u)_n}{(1+u)_n}, \qquad \mathcal{G}(u,t) = \frac{\Gamma(1+u)^2\Gamma((1+t)/2)\Gamma((1-t)/2)}{\Gamma((1+t)/2+u)\Gamma((1-t)/2+u)}.$$

Theorem 1.2 (Harmonic composition generator)**.** *For $|t| < 1$ and $\operatorname{Re} u > -1$,*

$$\sum_{n\geq 0} \frac{t^2 P_n(u)}{(2n+1)((2n+1)^2 - t^2)} = \frac{\mathcal{G}(u,t) - \mathcal{G}(u,0)}{4u},$$

with a removable value at $u = 0$. If $C_m(n) = \sum_{\mathbf{r}\vDash m} 2^{\ell(\mathbf{r})} H_n(\mathbf{r})$ uses strictly decreasing finite harmonic indices and $C_0(n) = 1$, then

$$\sum_{n\geq 0} \frac{C_m(n)}{(2n+1)^{2k+1}} = \frac{(-1)^m}{4}[u^{m+1}t^{2k}]\mathcal{G}(u,t), \qquad m \geq 0,\ k \geq 1.$$

All these values lie in $\mathbb{Q}[\log 2, \zeta(2), \zeta(3), \ldots]$.

Proof. In the normalized Rogers–Dougall summand take center $1/2$, parameters $b = (1+t)/2$, $c = (1-t)/2$, and $d_0 = 1$, subtract the instance $t = 0$, and multiply by $\Gamma(1+u)/\Gamma(1-u)$. This gives the first identity initially in $0 < \operatorname{Re} u < 1$. Its summands are $O(n^{-3-2\operatorname{Re} u})$ locally uniformly; the finite product P_n is holomorphic in $\operatorname{Re} u > -1$. Thus analytic continuation and Cauchy's formula give the stated domain and every fixed coefficient interchange. The factor $4u$ includes the change of center $n + 1/2 = (2n+1)/2$.

Expand each finite factor as $(1-u/j)/(1+u/j) = 1 + 2\sum_{r\geq 1}(-u/j)^r$ to obtain $P_n(u) = \sum_m (-1)^m C_m(n) u^m$. Coefficient extraction proves the second identity. Finally

$$\log \mathcal{G}(u,t) = 4u\log 2 + \sum_{r\geq 2} \frac{2(-1)^r(2-2^r)}{r}\zeta(r)u^r + \sum_{r,j\geq 1} \frac{2(-1)^{r+1}}{r}\binom{r+2j-1}{2j}(2^r - 2^{-2j})\zeta(r+2j)u^r t^{2j},$$

by the half-argument polygamma values. Only finitely many terms are needed for each requested coefficient. This proves the algebra claim, not independence of its generators. □

For example, with $L = \log 2$,

$$\sum_{n\geq 0} \frac{H_n^2}{(2n+1)^3} = \frac{31}{8}\zeta(5) - \frac{45}{8}L\zeta(4) + \frac{7}{2}L^2\zeta(3) - \frac{7}{8}\zeta(2)\zeta(3),$$

$$\sum_{n\geq 0} \frac{2H_n^3 + H_n^{(3)}}{(2n+1)^3} = -14L^3\zeta(3) + \frac{21}{2}L\zeta(2)\zeta(3) - \frac{21}{4}\zeta(3)^2 + \frac{135}{4}L^2\zeta(4) - \frac{93}{2}L\zeta(5) + \frac{945}{64}\zeta(6).$$

The full report supplies the general-center polygamma moments, all Stieltjes boundary jets, colored-polylogarithmic and integral forms, Euler primitives, a quartic example, and reproducible diagnostics. Classical summation and the earlier all-resonance continuation remain separately attributed. Global priority and numerical period independence are not claimed.